PT MERDEKA TSINGSHAN INDONESIA

STANDARD OPERATING PROCEDURE
ON THE GRATE BED POST
上篦床岗位

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1. Purpose 目的

This procedure covers the operator's responsibility for visual inspection, start up and shutdown of on the grate bed post under complete closure.

此程序涵盖操作员对完全关闭下的上篦床岗位进行目视操作前检查、启动运行和停车的责任。

2. Functional Description 功能说明

It is the responsibility of the operator to ensure that on the grate bed post is visually inspected before and after each shift to determine whether it can start and operate safely, or whether other activities (such as maintenance and repair or other operator tasks) must be performed before starting, to determine the sequential operation method of on the grate bed post and smooth operation, Stop order of normal stops and emergency stops.

操作员有责任确保在从停机启动之前和每个班次开始时对上篦床岗位进行目视检查，确定是否可以启动并安全运行，或者是否必须在开始启动之前执行其他活动(例如维护修理或其他操作员任务)，确定上篦床岗位的按照顺序启动的操作方法及平稳运行，正常停车和紧急停车的停止顺序。

3. References 参考

- MTI-CP-SOP-Post Technical Operation Procedures. MTI-CP-SOP 岗位操作规程
- Distributing Vehicle (4303-BCV-003). 摆动胶带机 (4303-BCV-003)
- Roller Screening Machine (4303-SCM-001/002). 辊式筛分机 (4303-SCM-001/002)
- Wide Belt Conveyor (4303-BCV-004). 宽胶带机 (4303-BCV-004)
- Hot Blast Stove (4303-FUR-001/002/003/004). 热风炉 (4303-FUR-001/002/003/004)
- Chain Grate (4303-CHG-001). 链篦机 (4303-CHG-001)
- Temperature Regulating Fan (4303-BLO-002/004/006/008). 调温风机 (4303-BLO-002/004/006/008)
- Combustion Supporting (4303-BLO-001/003/005/007). 助燃风机 (4303-BLO-001/003/005/007)
- Heat Recovery Fan (4303-FAN-001/002/003/004). 循环风机 (4303-FAN-001/002/003/004)
- Vibrating Screen (4303-SCM-003). 振动筛 (4303-SCM-003)

4. PPE 个人保护装备



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防尘口罩　防毒口罩

5. Hazards & Control 危害与控制

Description of Hazard / Environmental Issue 危害	Control Strategy 控制
1. Mechanical injury; 2. Dust; 3. Electric shock; 4. Slippery conditions and a trip to the danger; 5. Asphyxia; 6. Empyrosis, poisoning; 7. Object falling from high altitudes; 8. Noise. 1. 机械伤害； 2. 灰尘； 3. 触电； 4. 湿滑的条件，绊倒危险； 5. 窒息； 6. 烫伤，中毒； 7. 高空坠物； 8. 噪音。	1. Install a protective cover on the rotating part of the equipment; 2. Wear dust mask as required; 3. Do not contact the electrical switch with wet hands during operation, Regular inspection of the equipment leakage protection device is complete, effective; 4. Please hold the handrail up and down the stairs; 5. Clear clearance or maintenance operations; 6. Wear good labor protection goods; 7. Wear labor protection goods and check the safety of stacking items on the operation platform; 8. Wear the earplugs to protect your hearing. 1. 设备的转动部位安装防护罩； 2. 按要求佩戴防尘口罩； 3. 操作时不要湿手接触电器开关，定期检查设备漏电保护装置齐全、有效； 4. 上下楼梯请扶好扶手； 5. 清仓或维修作业执行审批程序； 6. 穿戴好劳保护品； 7. 穿戴好劳保护品，检查各平台的物品堆放安全性； 8. 佩戴好耳塞，保护听力。

6. Equipment 工具/ 设备

4303-LUS-001	Dry Oil Lubrication干油润滑系统
4303-REG-001 ~ 028	Bellows Regulating Valve风箱调节阀
4303-EFV-001	Electrical Flashboard Valve电动插板阀
	Clean the site tools清扫现场工具

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7. Procedure 程序

№	Stage	Key Points
1	General Check 普查	1. Confirm the pre-job technical training; 2. Wear suitable labor protection products, check and confirm that there is no unsafe state or safety risks on the site; 3. Check and clean the equipment and the site sanitation; 4. Contact with the process before and after, and give clear instructions. 1. 确认经过岗前技术培训; 2. 穿戴好适合的劳保护品，检查确认现场没有存在不安全状态或安全隐患; 3. 检查并打扫设备、现场卫生; 4. 与前后工序取得联系，并发出明确指令。
2	Inspect Distributing Vehicle (4303-BCV-003) 检查摆动胶带机 (4303-BCV-003)	1. The lubrication and sealing of each bearing is good, the reducer oil level is appropriate, the upper and lower roll lubrication is good, flexible operation; 2. Check whether each part of the conveyor is intact and whether the tension of the conveyor belt is appropriate; whether the corresponding belt tightening device and dustproof facilities are in good condition; 3. Check whether there is a firm and reliable protective cover at the pulley or the coupling of the transmission part; 4. Check whether the installed electrical facilities and signal devices are in good condition and responsive; 5. Connect the power supply, start several times, if the operation is normal, the conveyor can run continuously; 6. Observe the swing range of the swing head. In the normal trace range, the belt does not deviate; 7. During the operation process, check whether the tape machine emergency stop device, belt deviation correction and speed measuring sensor are in good condition. 1. 各部轴承润滑与封闭良好，减速机油位适当，上下托滚润滑良好，运转灵活; 2. 检查输送机各部分是否完好无损，输送带的张紧度是否合适；相应的皮带拉紧装置及防尘设施是否完好; 3. 检查传动部分的联轴器处或皮带轮处是否有牢固可靠的防护罩; 4. 检查装设的电器设施及信号装置是否完好和反应灵敏; 5. 接通电源，点动开车几次，如运转正常，即可使输送机连续运转; 6. 观察摆头的摆动幅度，在正常行迹区间，皮带无跑偏

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№	Stage	Key Points
		现象； 7. 运转过程中，检查胶带机急停装置、皮带纠偏和测速传感器是否控制灵敏。
3	Inspect Roller Screening Machine(4303-SCM-001/002) 检查辊式筛分机(4303-SCM-001/002)	1. Check the hydraulic drive assembly for loose or damaged parts, oil leakage or other damage; 2. Check the smooth roller of the sieve surface, the fastening of each bolt and the damage of the screen surface, and check whether there is any foreign body on the roller distributor; 3. Check whether the roller starts smoothly and quickly, has no special noise, whether the pendulum is produced, and each sieve roller rotates flexibly without jam; 4. Check whether the feeder discharge plate is loose or damaged; 5. Check whether the side panel components on both sides of the feeder are worn; 6. Ensure that the safety net is in place and safe. 1. 检查液压驱动组件是否有松动或损坏的零件、漏油或其他损坏； 2. 检查筛面辊子光滑情况、各部螺栓紧固情况和筛面破损情况，检查辊式布料器上面是否有异物； 3. 检查辊子是否启动平稳迅速，无特殊噪音，是否产生横摆，各筛辊转动灵活； 4. 检查进料器排料板是否松动或破损； 5. 检查进料器两侧的侧板组件是否磨损； 6. 确保安全网是否到位并安全。
4	Inspect Wide Belt Conveyor(4303-BCV-004) 检查宽胶带机(4303-BCV-004)	1. Each bearing is well lubrication and closed, the reducer oil level is appropriate, the upper and lower roller lubrication is good, the operation is flexible, whether the wear of the first and tail wheels and the tape are filled with materials, and whether the cleaning device of the first and tail wheels is in good condition; 2. Comprehensively check whether all parts of the belt conveyors are intact, whether the tension of the tape is appropriate, and whether the corresponding tape tightening device and dust-proof facilities are in good condition; 3. Whether the electrical facilities and control system are intact, whether the protective device is firm, and whether the funnel baffle and liner are damaged; 4. Check whether the cushion rollers of the impact load parts are intact; 5. Check the installation of belt sweeper; 6. Whether the installed electrical facilities and signal

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№	Stage	Key Points
		devices are intact and responsive; 7. Place the field control box in the manual position, press the start button, slowly rotate the potentiometer, and watch whether the running direction of the tape is correct, and whether the belt can run away; 8. After checking and contacting the post, press the start button; 9. During the operation process, check whether the tape machine emergency stop device, belt deviation correction and speed measuring sensor are in good condition. 1. 各部轴承润滑与封闭良好，减速机油位适当，上下托滚润滑良好，运转灵活，首、尾轮磨损情况与胶带间是否夹有物料，首、尾轮的清扫装置是否完好； 2. 全面检查宽胶带机各部分是否完好无损，胶带的张紧度是否合适，相应的胶带拉紧装置及防尘设施是否完好； 3. 电器设施和控制系统是否完好，防护装置是否牢固，漏斗挡板和衬板是否损坏； 4. 检查承受冲击载荷部位的缓冲托辊是否完好； 5. 检查宽胶带机清扫器安装是否正常； 6. 装设的电器设施及信号装置是否完好和反应灵敏； 7. 将现场控制盒置于手动位置，按一下启动按钮，迟缓旋动电位器，观看宽胶带运转方向能否正确，皮带是否跑偏； 8. 检查无误并与前后岗位联系后，按下启动按钮； 9. 运转过程中，检查胶带机急停装置、皮带纠偏和测速传感器是否控制灵敏。
5	Inspect Hot Blast Stove(4303-FUR-001/002/003/004) 检查热风炉(4303-FUR-001/002/003/004)	1. Check the electrical control cabinet in standby state before ignition; 2. Check the supply pressure to meet the safety requirements, check whether the valves and pipes leak and the valves are closed; 3. Check whether the motor steering, combustion aid fan and flue gas induced draft fan are correct, and check whether the wind is flexible, and the air valve is closed; 4. Check whether the compressed air pressure of the point torch is normal. 1. 点火前要先检查电气控制柜处于待机状态； 2. 认真检查供气压满足安全要求，检查各阀门和管道有没有渗漏情况，以及阀门是否处于关闭状态，如有问

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№	Stage	Key Points
		题需要及时处理； 3. 检查电机转向、助燃风机、烟气引风机是否正确，并检查调节风是否灵活，并且要将风门调制到关的状态； 4. 检查点火枪压缩空气气压是否正常。
6	Inspect Chain Grate (4303-CHG-001) 检查链篦机 (4303-CHG-001)	1. Check the upper and lower return cleaning of the grate bed, and not any foreign matter is allowed; 2. Check all equipment, valves, pipes in all jurisdiction; 3. Check whether the grate steering is correct, check the electrical inspection and test run, whether the air circuit and circuit system are normal; 4. Check whether the mechanical overload protection micro-switch of the main reducer is effective; 5. Check the welding of the ring at both ends of the small shaft of the grate bed; 6. Check whether the head sprocket, tail sprocket and top sprocket are well engaged; 7. Check whether the tension degree of the grate bed is appropriate and whether the tension of each chain is equal; 8. Check whether the instruments are complete and the debugging is normal; 9. Check whether the lubrication point has fully supplied fat and confirm that the lubrication system is normal, and check that there is no leakage in the lubrication point; 10. Check whether the gap between the shovel plate, the roller and the grate bed is adjusted in place, and the gap on the lower return side is normal; 11. Check whether the body tension device of the grate machine is normal; 12. Check whether the grate plate reset device is normal; 13. Check whether the installation of large scraper is normal and whether the use is flexible. 1. 检查链篦机篦床的上面和下回程清理干净，不允许有任何异物存在； 2. 检查所有管辖范围内热风管道控制阀门是否灵活； 3. 检查链篦机转向是否正确，检查进行电气检查和试运转，气路、电路系统是否正常； 4. 检查主减速机的机械过载保护的微动开关是否有效； 5. 检查篦床小轴两端卡环焊接完好； 6. 检查头部链轮、尾部链轮、上托链轮是否啮合良好；

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№	Stage	Key Points
		7. 检查链篦床的张紧程度是否合适，各排链子张紧是否相等； 8. 检查各处仪表是否齐全，并且调试正常； 9. 检查各润滑点是否已经充分供脂并确认润滑系统正常，同时检查各润滑点有无泄漏； 10. 检查主减速机油位是否正常； 11. 检查链篦机机体拉紧装置是否正常； 12. 检查篦板复位装置是否正常； 13. 检查出料大刮板安装是否正常，使用是否灵活。
7	Inspect Temperature Regulating Fan(4303-BLO-002/004/006/008),Combustion Supporting(4303-BLO-001/003/005/007),Heat Recovery Fan(4303-FAN-001/002/003/004) 检查调温风机(4303-BLO-002/004/006/008),助燃风机(4303-BLO-001/003/005/007),循环风机(4303-FAN-001/002/003/004)	1. Manual turning 2~3 turn, flexible rotation, no scraping, stuck and other abnormal phenomena; 2. The fan air outlet valve is normal, and the fan inlet valve is closed; 3. Close the power switch, first click to confirm the correct steering, no noise and other abnormal conditions; 4. After the fan starts and runs, observe the change of the fan sound, check the vibration and noise of the fan bearing and motor bearing, and check the vibration of the fan (including impeller and connector); 5. Check whether the lubricating oil level of the fan is normal. 1. 手动盘车2～3转，转动灵活，不得有刮蹭、卡住等异常现象； 2. 风机出风阀门正常，风机进口阀门关闭； 3. 合上电源开关，先点动确认转向正确、无杂音等异常情况； 4. 风机启动运行后，观察风机声音的变化，检查风机轴承、电机轴承的振动和噪声，检查风机振动(包括叶轮、连轴器)； 5. 检查风机润滑油位是否正常。
8	Inspect Vibrating Screen(4303-SCM-003) 检查振动筛(4303-SCM-003)	1. Check the oil level in the vibrator, check the screen surface tension, the bolt tightening of each part and the screen surface damage; 2. Check the working condition of the shock shock and the sieve box, observe whether the sieve starts smoothly and quickly, whether the vibration and operation are stable, there is no special noise, whether the amplitude meets the requirements, and the vibration of the sieve box should not produce yaw； 3. Check whether the vibration spring around the vibrating screen is normal and there is no broken spring

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		phenomenon; 4. Check whether the sieve pin around the vibrating screen is normal. 1. 检查振动器中的油位情况，检查筛面张紧情况、各部螺栓紧固情况和筛面破损情况； 2. 检查激振器和筛箱工作情况，观察筛子是否启动平稳迅速，振动和运行是否稳定，无特殊噪音，振幅是否符合要求，筛箱振动不应产生横摆； 3. 检查振动筛四周振动弹簧是否正常，无断簧现象； 4. 检查振动筛四周筛体稳定销是否正常。
9	Check the other Equipment and Facilities 检查其它设备设施	1. Dry Oil Lubrication(4303-LUS-001) a. Check whether the inlet and outlet pipes and auxiliary fittings of lubricating oil meet the requirements and whether oil leakage; b. Check whether the lubricating oil control components is normal; c. Check whether the oil pump is normal. 1. 干油润滑系统(4303-LUS-001) a. 检查各润滑油出入口管路及附属管件是否符合要求，是否漏油; b. 检查润滑油控制组件是否正常； c. 检查油泵是否正常。 2. Bellows Regulating Valve(4303-REG-001～028),Electrical Flashboard Valve(4303-EFV-001) a. Check whether the control valve is normal; b. Check whether the regulating valve is flexible or normal; c. Whether the regulating valve is in a closed state. 2. 风箱调节阀(4303-REG-001～028),电动插板阀(4303-EFV-001) a. 检查调节阀控制组件是否正常； b. 检查调节阀是否灵活，是否正常； c. 调节阀是否处于关闭状态。 3. Swing tape machine (4303-BCV-003) (4303-CHE-001) Size ball chute (4303-CHE-002/003) Guide plate (4303-CHE-004), vibration sieve (4303-SCM-003) (4303-CHE-005/006,4303-CHE-007) a. Check whether the feeding chute is in good condition; b. Check whether the blocking detection switch of the

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№	Stage	Key Points
		feeding chute is normal. 3. 摆动胶带机(4303-BCV-003)下料溜槽(4303-CHE-001),辊式筛分机(4303-SCM-001/002)大小球溜槽(4303-CHE-002/003),导料板(4303-CHE-004),振动筛(4303-SCM-003)排料溜槽,进料溜槽(4303-CHE-005/006,4303-CHE-007) a. 检查各下料溜槽是否完好; b. 检查下料溜槽的堵料检测开关是否正常。 4. Temperature regulating fan (4303-BLO-002/004/006/008) auxiliary combustion fan (4303-BLO-001/003/005/007) inlet butterfly valve,Heat Recovery Fan(4303-FAN-001/002/003/004)Inlet valve (32GB) a. Flexible open, temporarily closed state. 4. 调温风机(4303-BLO-002/004/006/008)助燃风机(4303-BLO-001/003/005/007)进口蝶阀,循环风机(4303-FAN-001/002/003/004)进口阀门(32GB) a. 开启灵活，暂时处于关闭状态。
10	Inspect Materials Level 检查物料料位	1. Check whether the material is stored on the swing tape machine (4303-BCV-003), and there is no adhesive deposit on the surface of the head and tail wheels and the junction of the belt; 2. Check whether there is material accumulation between the roller sieve machine (4303-SCM-001/002) and the discharge plate and the main structure to ensure that there is no accumulation in the discharge chute; 3. Check whether the material is stored on the wide tape machine (4303-BCV-004), and there is no adhesive deposit on the surface of the first and tail wheels and the joint of the belt; 4. Check whether the gas inlet pressure of the hot air furnace (4303-FUR-001/002/003/004) is about 25kPa, The air inlet wind pressure of the combustion nozzle is 2~3kPa; 5. Check whether the material is stored on the grate (4303-CHG-001); 6. Check for the vibrating screen (4303-SCM-003); 7. Check whether the lubricating oil in the dry oil lubrication system (4303-LUS-001) is sufficient. 1. 检查摆动胶带机(4303-BCV-003)上是否存料，首尾轮表面与皮带结合部无黏附积矿存在;

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№	Stage	Key Points
		2. 检查辊式筛分机(4303-SCM-001/002)排料板和主体结构之间是否有物料堆积，确保排料溜槽没有堆积物； 3. 检查宽胶带机(4303-BCV-004)上是否存料，首尾轮表面与皮带结合部无黏附积矿存在； 4. 检查热风炉(4303-FUR-001/002/003/004)燃气进口压力是否在25kPa左右，燃烧嘴空气进口风压2～3kPa； 5. 检查检查链篦机 (4303-CHG-001)上是否存料； 6. 检查振动筛(4303-SCM-003)上是否存料； 7. 检查干油润滑系统(4303-LUS-001)内润滑油是否足够。
11	Start-Up Of on the Grate Bed Post 上篦床岗位开机	1. No-load start should be guaranteed when starting; 2. The driving sequence is along the material direction, from back to front that is the reverse drive, Each equipment needs to be switched on later; At the same time, Confirm that the process behind the grate bed post has been started and has the feeding conditions: Open the vibrating screen (4303-SCM-003) ⟹ Open the electric plug panel valve (4303-EFV-001) ⟹ Open the dry oil lubrication system (4303-CHG-001) for the first time (4303-LUS-001) (periodically open the dry oil lubrication system), and then open the grate (4303-CHG-001), and bake the new grate before no-load test and production. The heat source used for baking generally has no special requirements, but it must be able to control the heating speed to avoid local overheating. According to the baking curve gradually warming, insulation, heating, insulation, sequence into the normal production state. Use gas hot air to bake, control the temperature and maintain enough time to ensure the water vapor in the oven. The specific oven curve refers to the oven scheme provided by the resistant material manufacturer; run slowly from low speed to high speed, the running speed is 0.5~0.9m/min, pay attention to the grate dryer is strictly prohibited to reverse at any time ⟹ Before ignition (4303-FUR-001/002/003/004), open the fan inlet butterfly valve (350BF), open the fan (4303-FAN-005), Open the inlet valve (32GB), open the circulation fan (4303-FAN-001/002/003/004), Open the inlet valve (32GB), open

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№	Stage	Key Points
		the auxiliary combustion fan (4303-BLO-001/003/005/007), The fan operation frequency is 5~10Hz; Then open the hot air furnace (4303-FUR-001/002/003/004), Close the gas air inlet small ball valve first, Press the "gas turbine / mother fire start" button, After 10 seconds of the work indicator light, Check whether the ignition of the lighter gun is normal from the rear viewing hole of the furnace, The lighter gun fires normally, After opening the inlet air gate valve to 50% opening and the gas inlet ball valve to about 30%, Press the Big Fire " button, System entry program automatic control, Ignition transformer working ignition, Ignition solenoid valve is open; After the ignition is running normally, After opening the inlet valve (32GB) and opening the temperature regulating fan (4303-BLO-002/004/006/008), First fill the gas and then increase the frequency of each fan, The pressure change of the air guide will adjusts the opening of the air / combustion proportional valve, Can achieve the automatic ratio of air and gas, Until the flame is in main blue and slightly red⟹Open the bellows regulating valve (4303-REG-001~028) ⟹open roller sieve machine (fine) (4303-SCM-002)⟹open wide tape machine (4303-BCV-004) ⟹Open roller sieve machine (coarse) (4303-SCM-001)⟹ open swing tape machine (4303-BCV-003). 1. 启动时应保证空载启动； 2. 开车顺序为沿物料走向，从后往前开即逆向开车，各设备都需要延时开启；**同时，确认上篦床岗位后面的工序已经开车并具备给料条件：** **开启振动**筛(4303-SCM-003)⟹**开启**电动插板阀(4303-EFV-001)⟹**第一次**开启链篦机（4303-CHG-001)**时先开启**干油润滑系统(4303-LUS-001)**(以后周期性开启**干油润滑系统)，**然后**开启链篦机（4303-CHG-001)，新投入使用的链篦机无负荷试车之后投料生产之前，需要进行烘烤。烘烤所用热源一般无特殊要求，但必须能够控制升温速度，避免局部过热。按烘烤曲线逐步升温、保温、升温、保温，顺次进入正常生产状态。利用煤气热风炉热风进行烘烤，控制温度、维持足够时间，保证炉内水汽彻底排出，具体烘炉曲线参照耐材厂家提供的烘炉方案；运行时从低速慢慢运行到高速，运行速度为0.5 ~ 0.9m/min，注意在任何时候严禁链篦烘干机开倒车

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№	Stage	Key Points
		⟹热风炉(4303-FUR-001/002/003/004)点火前先打开风机进口蝶阀(350BF)、开启风机(4303-FAN-005)，打开进口阀门(32GB)、开启循环风机(4303-FAN-001/002/003/004)，打开进口阀门(32GB)、开启助燃风机(4303-BLO-001/003/005/007)，风机运转频率5～10Hz；然后开启热风炉(4303-FUR-001/002/003/004)，先关闭燃气进气口小球阀，按下“燃机/母火启动”按钮，工作指示灯亮起10秒钟以后，从炉后部的观火孔查看点火枪点火是否正常，点火枪点火正常，打开进口空气闸阀调到50%开度和燃气进气口小球阀到30%左右后，按下“大火”按钮，系统进入程序自动检控，点火变压器工作点火，点火电磁阀打开；点火运行正常后，打开进口阀门(32GB)、开启调温风机(4303-BLO-002/004/006/008)后，先加气后加大各风机的频率，空气导压管的压力变化会调节空/燃比例阀的开度，可以实现空气、燃气自动配比，直至火焰呈现主蓝加微红色⟹开启风箱调节阀(4303-REG-001～028)⟹开启辊式筛分机(细料)(4303-SCM-002)⟹开启宽胶带机(4303-BCV-004)⟹开启辊式筛分机(粗料)(4303-SCM-001)⟹开启摆动胶带机(4303-BCV-003)。
12	On The Grate Bed Post Stops 上篦床岗位停机	1. Orderly shutdown a. The materials in the standby machine should be discharged before stopping; b. Stop in the reverse order of driving order, Each equipment needs to stop: Stop swing tape machine (4303-BCV-003)⟹Stop roller screen machine (coarse material) (4303-SCM-001)⟹Stop wide tape machine (4303-BCV-004)⟹Stop roller screen machine (fine material) (4303-SCM-002)⟹ Close the bellows control valve (4303-REG-001~028)⟹Stop the hot air furnace (4303-FUR-001/002/003/004), press the stop button, stop the combustion system and close the valve on the gas pipeline⟹ stop the combustion support fan (4303-BLO-001/003/005/007) and close the inlet valve (32GB)⟹ Stop the temperature regulating fan (4303-BLO-002/004/006/008) and close the inlet valve (32GB)⟹Stop the circulation fan (4303-FAN-001/002/003/004) and

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№	Stage	Key Points
		close the inlet valve (32GB)⟹Stop grate (4303-CHG-001), slowly reduce the speed from high speed operation, when the internal temperature of the hood decreases to 100℃, then close the switch to stop running; During the cooling process, pay attention to the detection of the bellows and hood temperature to prevent temperature rise and sharp temperature drop, and the cooling speed of the hood temperature should be controlled within 120℃/h, Close the dry oil lubrication system (4303-LUS-001)⟹close the electric panel valve (4303-EFV-001)⟹ Stop the vibrating screen (4303-SCM-003)⟹Clean the site for hygiene. 1. 正常停车 a. 停车时应待机内物料排净后再停车; b. 按照开车顺序逆序停车，各设备都需要延时停止： 停止摆动胶带机(4303-BCV-003)⟹停止辊式筛分机(粗料)(4303-SCM-001)⟹停止宽胶带机(4303-BCV-004)⟹停止辊式筛分机(细料)(4303-SCM-002)⟹关闭风箱调节阀(4303-REG-001～028)⟹停止热风炉(4303-FUR-001/002/003/004)，按下停止按钮，燃烧系统停止工作，关闭燃气管道上阀门⟹停止助燃风机(4303-BLO-001/003/005/007)，关闭进口阀门(32GB)⟹停止调温风机(4303-BLO-002/004/006/008)，关闭进口阀门(32GB)⟹停止循环风机(4303-FAN-001/002/003/004)，关闭进口阀门(32GB)⟹停止链篦机 (4303-CHG-001)，由高速运行慢慢降低速度，当罩体内部温度降低至100℃，再关闭开关停止运行；降温过程中注意检测风箱和上罩温度，防止温度升高和温度急降，上罩温度降温速度控制在 120℃/h 之内，关闭干油润滑系统(4303-LUS-001)⟹关闭电动插板阀(4303-EFV-001)⟹停止振动筛(4303-SCM-003)⟹打扫现场卫生。 2. Emergency Stop a. When an emergency occurs, Close the swing tape machine (4303-BCV-003), Roller Screen Machine (coarse) (4303-SCM-001), Wide adhesive tape machine (4303-BCV-004), Roller-screen machine (fine material) (4303-SCM-002), Tier machine (4303-

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№	Stage	Key Points
		CHG-001), Return heat fan (4303-FAN-001/002/003/004), Temperature fan (4303-BLO-002/004/006/008), Comradler (4303-BLO-001/003/005/007), Hot-air Furnace (4303-FUR-001/002/003/004), Electric power panel valve (4303-EFV-001), Stop the vibration screen (4303-SCM-003), After all stop the operation, Check whether the start / stop button of each device is in the stop state, Close return fan (4303-FAN-001/002/003/004), temperature fan (4303-BLO-002/004/006/008), combustion fan (4303-BLO-001/003/005/007) inlet valve (32GB), at the same time, the upper level of equipment all need to stop; b. After the accident, the equipment is restored to normal. If production still needs to be conducted, and the operation shall be restarted in the driving sequence; c. After the accident treatment, the equipment returns to normal. If the production is not necessary, operate according to the normal parking order. 2. 紧急停车 a. 当出现紧急事故，关闭摆动胶带机(4303-BCV-003),辊式筛分机(粗料)(4303-SCM-001),宽胶带机(4303-BCV-004),辊式筛分机(细料)(4303-SCM-002),链篦机 (4303-CHG-001),循环风机(4303-FAN-001/002/003/004),调温风机(4303-BLO-002/004/006/008),助燃风机(4303-BLO-001/003/005/007),热风炉(4303-FUR-001/002/003/004),电动插板阀(4303-EFV-001),停止振动筛(4303-SCM-003)，全部停止运行后，检查各设备的启动/停止按钮是否处于停止状态，关闭循环风机(4303-FAN-001/002/003/004)、调温风机(4303-BLO-002/004/006/008)、助燃风机(4303-BLO-001/003/005/007)进口阀门(32GB)，同时，上一级设备均需停车； b. 事故处理后，设备恢复正常，如仍需要进行生产，按照开车顺序重新启动操作； c. 事故处理后，设备恢复正常，如不需要进行生产，按照正常停车顺序进行操作。

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№	Stage	Key Points
13	Permits ＆Isolation Sheets 许可证和隔离表	1. Only authorized isolators can sign off and cancel permits and registration sheets. 1. 只有经过授权的隔离者才能签署和取消许可证和登记表。

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8. On The Grate Bed Post 上篦床岗位 Questionnaire

Declaration of Understanding

I (_____Learner_____) have completed the theory assessment and had all my valid, reliable, and consistent with my level of knowledge in the assessment task.

Signature
Date:

I (_____Assessor_____) have witnessed the above learner complete the theory assessment and believe the assessment has provided a valid, reliable, and consistent result.

Signature
Date:

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9. Appendix

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Standard Operating Procedure
标准操作程序

On the grate bed posts in the chloride plant
CP 上篦床岗位

SOP Details 程序明细

Processing Department 氯化厂	Procedure Level 程序级别: **3**	Revision Number 修订编号: 0
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Rev. No 版本号	Date 日期	Section 章节	Page 页码	Revision Description 修订说明	Reviewed By 审核人
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Approvals 批准

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